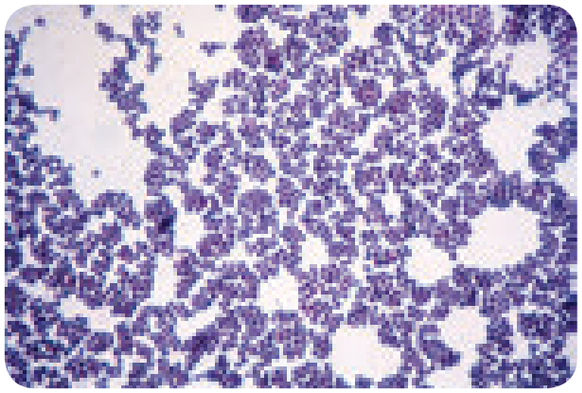


Figure 7.1: Gram staining of *Staphylococcus aureus*

7.3.3 Virulence Factors

1. Peptidoglycan → It is a polysaccharide polymer. It activates complement and induces the release of inflammatory cytokines.
2. Teichoic acid → it facilitates adhesion of cocci to the host cell surface.
3. Protein A → It is chemotactic, antiphagocytic, anticomplementary and induce platelet injury.
4. **Toxins:**
   a. Hemolysins – It is an exotoxin, those lysis red blood cells. They are of four types namely α-lysin, β-lysin, γ-lysin and delta lysin.
   b. Leucocidin – It damages PMNL (polymorphonuclear leucocytes) and macrophages.
   c. Enterotoxin – It is responsible for manifestations of Staphylococcus food poisoning.
   d. Exfoliative toxin – This toxin causes epidermal splitting resulting in blistering diseases.
   e. Toxic shock syndrome toxin – TSST is responsible for toxic shock syndrome.

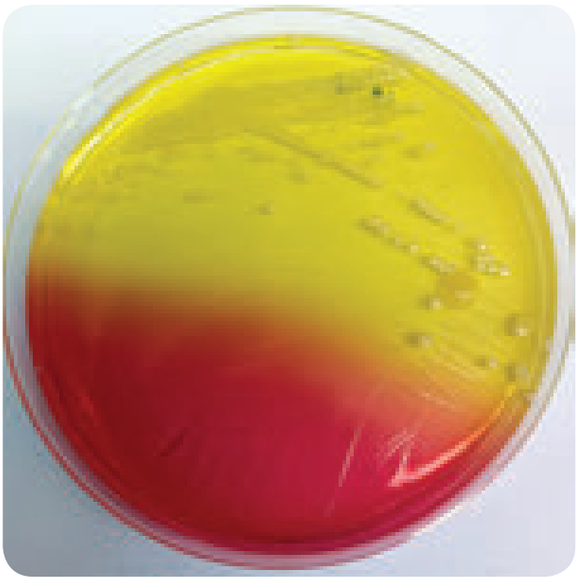


Figure 7.2: Colony morphology of *Staphylococcus aureus* on MSA

5. **Enzymes:** *S. aureus* produces several enzymes, which are related to virulence of the bacteria.
   a. Coagulase – It clots human plasma and converts fibrinogen into fibrin.
   b. Staphylokinase – It has fibrinolytic activity.
   c. Hyaluronidase – It hydrolyzes hyaluronic acid of connective tissue, thus facilitates the spread of the pathogens to adjacent cells.
   d. Other enzymes – S. *aureus* also produces lipase, nucleases and proteases

7.3.4 Pathogenicity

S. *aureus* is an opportunistic pathogen which causes infection most commonly at sites of lowered host resistance. (Example: damaged skin)

Mode of Transmission: *Staphylococcus* infections are transmitted by the following ways.

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